Exercise Sheet 3

Notation follows Lecture 3: $T^2 = \mathrm{U}(1) \times \mathrm{U}(1)$, $\varphi_\alpha(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$ and $L_\alpha = \varphi_\alpha(\mathbb{R})$. Coset spaces carry the quotient topology. A topological space is *compact* if every cover by open sets has a finite subcover. You may use the Heine–Borel theorem in $\mathbb{R}^n$, the fact that a continuous image of a compact space is compact, and the theorems cited in Lecture 3. Exercise 3.7 is an optional extension.

Exercise 3.1 (Closed subgroups of compact groups). Let X be a compact topological space.

(i) Prove that every closed subset $C \subseteq X$ is compact in the subspace topology.

(ii) Let K be a compact Lie group and $H \subseteq K$ a closed subgroup. Prove that H is a compact Lie group and that K/H is compact.

(iii) Let α be irrational. Give L_α the Lie group structure transported from $\mathbb{R}$ by φ_α. Is L_α compact in this structure? Explain why this does not contradict (ii).

Exercise 3.2 (Rational lines). Let $\alpha = p/q$, where p, q are integers, $q \geq 1$ and $\gcd(p, q) = 1$. This includes $\alpha = 0$, with $p = 0$ and $q = 1$.

(i) Prove that $\ker \varphi_\alpha = q\mathbb{Z}$.

(ii) Prove that $L_\alpha = \varphi_\alpha([0, q])$, and deduce that L_α is closed in T^2.

(iii) Define $\psi\colon \mathrm{U}(1) \to T^2$ by $\psi(z) = (z^q, z^p)$. Prove that ψ is an injective immersive homomorphism with image L_α.

Exercise 3.3 (The closure of an irrational line). Let G be a Lie group.

(i) Prove that the closure $\overline{H}$ of a subgroup $H \subseteq G$ is a subgroup.

(ii) Let α be irrational. Find $\overline{L_\alpha}$ and its Lie algebra. Compare it with $\mathrm{Lie}(L_\alpha) = \mathbb{R}(1, \alpha)$.

(iii) Let α be irrational and let $f\colon T^2 \to \mathbb{C}$ be continuous with $f(xl) = f(x)$ for all $x \in T^2$ and $l \in L_\alpha$. Prove that f is constant. Deduce that every continuous function from T^2/L_α to $\mathbb{C}$ is constant.

(iv) Let $\alpha = p/q$ as in Exercise 3.2. Show that $f(z, w) = z^{-p} w^q$ is a nonconstant continuous function with the invariance property of (iii).

Exercise 3.4 (Spheres as unitary coset spaces). Let $n \geq 2$. The group $\mathrm{U}(n)$ acts on $\mathbb{C}^n$ by matrix multiplication. The unit sphere $S^{2n-1} = \{v \in \mathbb{C}^n : |v| = 1\}$ has real dimension $2n - 1$. Use $\dim_{\mathbb{R}} \mathrm{U}(n) = n^2$ and $\dim_{\mathbb{R}} \mathrm{SU}(n) = n^2 - 1$.

(i) Prove that $\mathrm{U}(n)$ acts transitively on S^{2n-1}.

(ii) Prove that the stabiliser of e_n consists of the matrices $\mathrm{diag}(U', 1)$ with $U' \in \mathrm{U}(n-1)$. Check the dimension of $\mathrm{U}(n)/\mathrm{U}(n-1)$.

(iii) Prove that $\mathrm{SU}(n)$ also acts transitively, with stabiliser of e_n isomorphic to $\mathrm{SU}(n-1)$. For $n = 2$, compare the orbit map with the identification $\mathrm{SU}(2) \cong S^3$ of Lecture 1.

Exercise 3.5 (Isotropy and the kernel of an action). Use the space V, the coordinates $\xi(x, y, z)$, the sphere S, the base point N and the elements k_θ of Lecture 3.

(i) Prove that $U \in \mathrm{SU}(2)$ fixes every point of S exactly when $U = \pm I$.

(ii) Prove that the only $g \in \mathrm{SO}(3)$ fixing every point of S^2 is I.

(iii) Using the isotropy calculations of the lecture, list the elements of each circle stabiliser that act trivially on the tangent plane at the base point. Explain the consistency with (i) and (ii).

Exercise 3.6 (Normality and quotient groups). Recall $k_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ and $T = \{k_\theta\} \subseteq \mathrm{SU}(2)$.

(i) Prove that $\mathrm{SU}(n)$ is a closed normal subgroup of $\mathrm{U}(n)$, and that det induces a group isomorphism $\mathrm{U}(n)/\mathrm{SU}(n) \to \mathrm{U}(1)$. What do the cited quotient-group and homogeneous-space theorems add?

(ii) Let

$$U_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Show that $U_0 \in \mathrm{SU}(2)$ and compute $U_0 k_{\pi/2} U_0^{-1}$. Deduce that $\mathrm{SU}(2)/T$ has no coset group law.

(iii) Let B be the matrix of the lecture. Show that $B \in \mathrm{SU}(2)$ and $B k_\theta B^{-1} = k_{-\theta}$. Is B in T?

Exercise 3.7 (Optional: the upper half-plane). Let $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}\, z > 0\}$. The group $\mathrm{SL}_2(\mathbb{R})$ of real 2×2 matrices of determinant one is a closed subgroup of $\mathrm{GL}_2(\mathbb{R})$ of dimension three. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R})$, put

$$g \cdot z = \frac{az + b}{cz + d}.$$

(i) For $z \in \mathbb{H}$, prove that $cz + d \neq 0$ and $\mathrm{Im}(g \cdot z) = \mathrm{Im}\, z / |cz + d|^2$.

(ii) Prove that $(gh) \cdot z = g \cdot (h \cdot z)$.

(iii) Prove that the action is transitive and that the stabiliser of i is $\mathrm{SO}(2)$. Identify $\mathbb{H}$ as a coset space and check dimensions.